IP Internet protocol (location)
MAC Media access control (Identification)
} device

MAC

- Media access control
- It is a physical address ✓
- It is in hexa decimal format
- It contains 12 hexa decimal
- Each hexa contains 4 bit
- 12*4 = 48 bit length

NIC ~ MAC / 48 bit

S. Surendran

1C-C1-DE-33-8D-15

(Binary)

48 bit

24 bit vendor id

24 bit user id

IP

- Internet protocol
- It is a logical address
- It is in doted decimal format
- It contains 4 octet
- Each octet contains 8 bit
- 8*4 = 32 bit length
- IP = network+ host

NIC – IP

IPV4

IP Structure

32 bit

1st	2nd	3rd	4th
128 64 32 16 8 4 2 1	128 64 32 16 8 4 2 1	128 64 32 16 8 4 2 1	128 64 32 16 8 4 2 1
0 0 0 0 0 0 0 0	0 0 0 0 0 0 0 0	0 0 0 0 0 0 0 0	0 0 0 0 0 0 0 0
1 1 1 1 1 1 1 1	1 1 1 1 1 1 1 1	1 1 1 1 1 1 1 1	1 1 1 1 1 1 1 1

→ 0.0.0.0 Starting IP

→ 255.255.255.255 Ending IP

Class
A B C D E

Types of Communication

Unicast one - one

Multicast one - Group

Broadcast one - All

CLASS	RANGE	PORTION
A	0 - 127	N.H.H.H
B	128 - 191	N.N.H.H
C	192 - 223	N.N.N.H
D	224 - 239	Multicast
E	240 - 255	Broadcast R&D

} Invalid IP

80.0.0.10 → A → N.H.H.H

172.80.254.128 → B → N.N.H.H

192.255.72.82 → C → N.N.N.H

School

4th std

2 bit

2	1
0	0
0	1
1	0
1	1

3 bit

4	2	1
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1

5 bit

16 8 4 2 1

Pen ~ stop

Binary Combination

Pending

$\boxed{50}$. 0 . 0 . 10

N . H . H . H

Network = Street No

Host = Door No

$\boxed{50}$. 50 . 50 . 50 IP

172 . 182 . 210 . 32

N . N . H . H

192 .	192 .	192 .	192
N .	N .	N .	H

How many network octet?

3

How many host octet?

1

How many network bit?

24

How many host bit?

8

fresher

Job

90%

1 hour

(Note)

1 ->